

Network Security Assessment Report: SOITnetworkscan.xml

Date of Report: Friday, October 2, 2026, 16:39:05

Subject: Nmap Scan Analysis – SOITnetworkscan.xml

Scope: Internal Network Segment 10.200.108.0/24

1. Executive Summary

The scan of the 10.200.108.0/24 network reveals a complex environment comprising two distinct Active Directory domains (`soit.local` and `pcsupport.ac.nz`), Cisco network infrastructure, and a VMware virtualization layer. The environment contains several high-value targets, including Domain Controllers and an ESXi hypervisor. Notable security concerns include the use of unencrypted management protocols (Telnet, HTTP) and the presence of potentially misconfigured Cisco devices.

2. Key Findings & Attack Surface

A. High-Value Targets (Critical Infrastructure)

- **Domain Controllers (DCs):**
 - 10.200.106.8 (DC7.`soit.local`) and 10.200.108.41 (PFWDC01.`pcsupport.ac.nz`) are running Windows Server 2022. These hosts are the primary targets for credential harvesting and domain escalation via Kerberos, LDAP, and SMB.
- **Virtualization Layer:**
 - 10.200.108.44 (VMware ESXi) is active. Compromise of this host would allow for “guest-to-host” escapes and full control over all virtualized workloads.

B. Identified Vulnerabilities & Security Risks

- **Insecure Management Protocols:**
 - **Telnet (Port 23):** Detected on Cisco infrastructure (10.200.108.13, 10.200.108.15). Traffic, including administrative credentials, is transmitted in cleartext and is susceptible to sniffing.
 - **HTTP (Port 80):** Detected on Cisco devices, ESXi, and Linux hosts. Lack of TLS/SSL exposes management interfaces to Man-in-the-Middle (MitM) attacks.
- **Unnecessary Service Exposure:**
 - **Gnutella (Port 6346):** Detected on several Linux-based embedded devices (10.200.108.20-22). This protocol is non-standard for en-

terprise environments and may indicate unauthorized software or IoT vulnerabilities.

- **High Port Volatility:** Cisco devices exhibit an unusually high number of open ports in the 2000s and 6000s, suggesting potential service misconfigurations or undocumented backdoors.
- **Information Leakage:**
 - **SMB/RPC/LDAP:** Active exposure of these services across multiple domains provides an avenue for reconnaissance, enumeration of users, and potential exploitation of SMB-based vulnerabilities.

3. Asset Inventory Summary

Asset Type	Primary IP(s)	OS / Platform	Critical Services
Domain	10.200.108.8,	Windows Server	Kerberos, LDAP,
Controllers	10.200.108.41	2022	DNS, SMB
Hypervisor	10.200.108.44	VMware ESXi	SSH, VMware
			Auth, HTTP
Network	10.200.108.13,	Cisco IOS	Telnet, HTTP
Infrastructure	10.200.108.15		
Linux Servers	10.200.108.17,	Ubuntu / Debian	SSH, RPCbind
	10.200.108.45		
Embedded/IoT	10.200.108.20-22	Linux	lighttpd, Gnutilla
		(Embedded)	

4. Recommendations

1. **Disable Insecure Protocols:** Immediately transition all Cisco management from Telnet to SSH and replace HTTP with HTTPS.
2. **Hardening Hypervisors:** Restrict access to the VMware ESXi management interface (10.200.108.44) to a dedicated, isolated management VLAN.
3. **Service Minimization:** Audit and disable the Gnutella service on Linux hosts and investigate the high-range open ports on Cisco infrastructure.
4. **Segmentation Review:** Evaluate the necessity of the cross-domain visibility between `soit.local` and `pcsupport.ac.nz` to prevent lateral movement between domains.